Post Lab Tasks



Assignment Title: Post Lab Tasks
Subject Name: Programming Fundamentals Lab
Submitted by: Nabiha Shaukat
Roll Number: 26k-3166
Submitted to: Sir Shayan Ali
Date of Submission: 20 oct 2026

Question 1: Print the Following Pattern

Write a C program using nested loops to print the following pattern:

```
1
1 1
1 1 1
1 1 1 0
1 1 1 0 1
1 1 1 0 1 1
1 1 1 0 1 1 1
1 1 1 0 1 1 1 0
```

main.c — Run

```c
#include <stdio.h>
int main(){
    int i, j;

    for(i = 1; i <= 8; i++) {
        for(j = 1; j <= i; j++)  {
            if(j == 4 || j == 8)
                printf("0 ");
            else
                printf("1 ");
        }
        printf("\n");
    }
    return 0;
}
```

Output

```
1
1 1
1 1 1
1 1 1 0
1 1 1 0 1
1 1 1 0 1 1
1 1 1 0 1 1 1
1 1 1 0 1 1 1 0
```

Question 2: Print the Following Pattern

Write a C program using nested loops to print the following pattern:

```
1
1 0
1 0 1
1 0 1 0
1 0 1 0 1
```

main.c Run ▶

```c
#include <stdio.h>
int main(){
    int i, j;

    for (i = 1; i <= 5; i++) {
        for (j = 1; j <= i; j++) {
            if (j % 2 == 0)
                printf("0 ");
            else
                printf("1 ");
        }
        printf("\n");
    }
    return 0;
}
```

Output

```
1
1 0
1 0 1
1 0 1 0
1 0 1 0 1
```

Question 3: Matrix Addition (2D Arrays)

Write a C program to input two 3×3 matrices using separate 2D arrays, add their corresponding elements using nested loops, store the results in a third matrix, and display both input matrices and the resulting sum matrix.

main.c

Run ▶

```c
#include <stdio.h>
int main(){
    int A[3][3], B[3][3], Sum[3][3];
    int i, j;

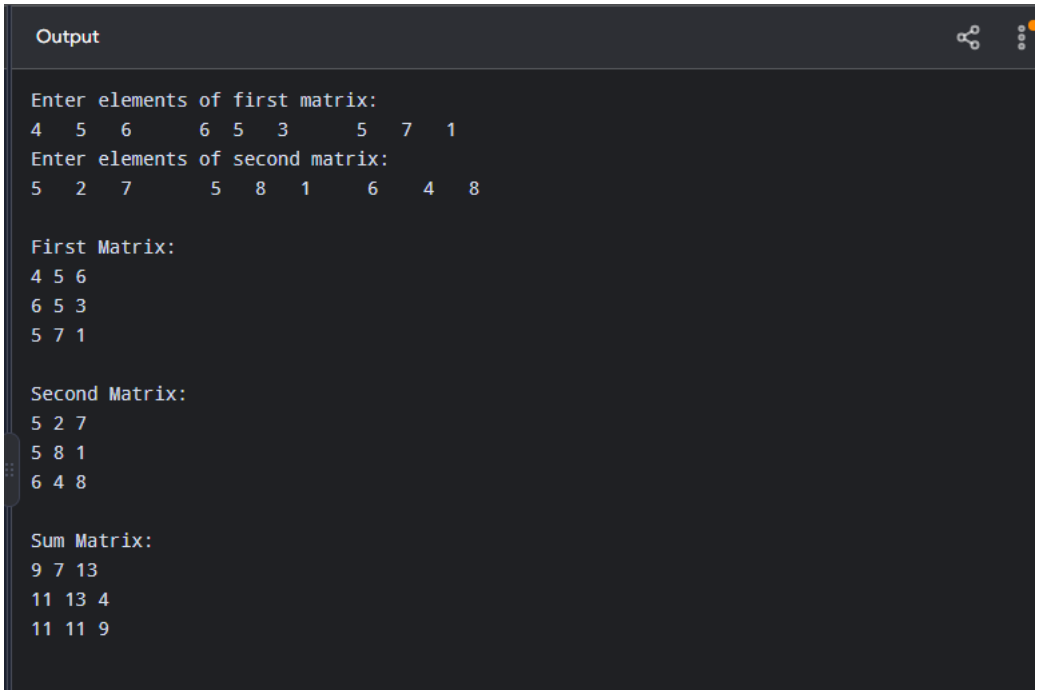
    printf("Enter elements of first matrix:\n");

    for (i = 0; i < 3; i++){
        for (j = 0; j < 3; j++) {
            scanf("%d", &A[i][j]);
        }
    }
    printf("Enter elements of second matrix:\n");

    for (i = 0; i < 3; i++){
        for (j = 0; j < 3; j++) {
            scanf("%d", &B[i][j]);
        }
    }
    for (i = 0; i < 3; i++)  {
        for (j = 0; j < 3; j++) {
            Sum[i][j] = A[i][j] + B[i][j];
        }
    }
    printf("\nFirst Matrix:\n");
    for (i = 0; i < 3; i++) {
        for (j = 0; j < 3; j++) {
            printf("%d ", A[i][j]);
        }
        printf("\n");
    }
    printf("\nSecond Matrix:\n");

    for (i = 0; i < 3; i++) {
        for (j = 0; j < 3; j++) {
            printf("%d ", B[i][j]);
        }
        printf("\n");
    }
    printf("\nSum Matrix:\n");

    for (i = 0; i < 3; i++) {
        for (j = 0; j < 3; j++) {
            printf("%d ", Sum[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```



Question 4: Student Marks Analysis (2D Array)

Write a C program using a 2D array to store the Mid and Final exam marks of 4 students (each row is a student, with one column for Mid and one for Final). Input the marks using nested loops, then calculate and display:

1. The total marks of each student (Mid + Final)
2. The highest marks in Mid, the highest marks in Final, and the highest marks overall across both
3. The lowest marks in Mid, the lowest marks in Final, and the lowest marks overall across both
4. The average of Mid marks and the average of Final marks, calculated separately

main.c

Run ▶

```c
#include <stdio.h>
int main(){
    int m[4][2];
    int i, j, total, hm, hf, ho, lm, lf, lo;
    int sm = 0, sf = 0;
    float am, af;

    for (i = 0; i < 4; i++) {
        printf("Enter Mid and Final marks of student %d: ", i + 1);

        for (j = 0; j < 2; j++)  {
            scanf("%d", &m[i][j]);
        }
    }
    hm = lm = m[0][0];
    hf = lf = m[0][1];
    ho = m[0][0];
    lo = m[0][0];

    for (i = 0; i < 4; i++){
        total = m[i][0] + m[i][1];
        printf("Total of student %d = %d\n", i + 1, total);

        sm = sm + m[i][0];
        sf = sf + m[i][1];

        if (m[i][0] > hm)
            hm = m[i][0];
        if (m[i][0] < lm)
```

main.c

Run ▶

```
29          if (m[i][0] < lm)
30              lm = m[i][0];
31          if (m[i][1] > hf)
32              hf = m[i][1];
33          if (m[i][1] < lf)
34              lf = m[i][1];
35
36          for (j = 0; j < 2; j++)  {
37              if (m[i][j] > ho)
38                  ho = m[i][j];
39
40              if (m[i][j] < lo)
41                  lo = m[i][j];
42          }
43      }
44      am = sm / 4.0;
45      af = sf / 4.0;
46
47      printf("\nHighest Mid = %d", hm);
48      printf("\nHighest Final = %d", hf);
49      printf("\nHighest Overall = %d", ho);
50      printf("\nLowest Mid = %d", lm);
51      printf("\nLowest Final = %d", lf);
52      printf("\nLowest Overall = %d", lo);
53      printf("\nAverage Mid = %.2f", am);
54      printf("\nAverage Final = %.2f\n", af);
55
56      return 0;
57  }
```

Output

```
Enter Mid and Final marks of student 1: 54  76
Enter Mid and Final marks of student 2: 72   87
Enter Mid and Final marks of student 3: 65   76
Enter Mid and Final marks of student 4: 76   86
Total of student 1 = 130
Total of student 2 = 159
Total of student 3 = 141
Total of student 4 = 162

Highest Mid = 76
Highest Final = 87
Highest Overall = 87
Lowest Mid = 54
Lowest Final = 76
Lowest Overall = 54
Average Mid = 66.75
Average Final = 81.25
```

Question 5: 3D Array Scenario (Paper Work)

A university stores the marks of students in a 3D array `int marks[2][3][3]`, where the first index is the semester (2 semesters), the second index is the student (3 students) and the third index is the subject (3 subjects). On paper, draw the array as 2 layers (one per semester) and label every cell with its index `marks[semester][student][subject]`. Decide what to insert: fill every cell with a realistic mark of your choice (0-100), writing the name of the
semester, each student and each subject on the margins of your diagram. Then, from your diagram, write down: (a) the index and value of the highest mark, (b) the marks of student 1 in semester 2 across all subjects, and (c) the total marks of each student in semester 1.

Question 5

marks [2][3][3]

2 = semester 1 and semester 2

3 = student 1, 2, 3

3 = subject 1, 2, 3

layer 1 – semester 1

student	subject 1	subject 2	subject 3
student 1	80	67	74
student 2	70	73	68
student 3	75	84	70

subject 1	subject 2	subject 3
marks [0][0][0] 80	marks [0][0][1] 67	marks [0][0][2] 74
marks [0][1][0] 70	marks [0][1][1] 73	mark [0][1][2] 68
marks [0][2][0] 75	marks [0][2][1] 84	marks [0][2][2] 70

layer 2 – semester 2

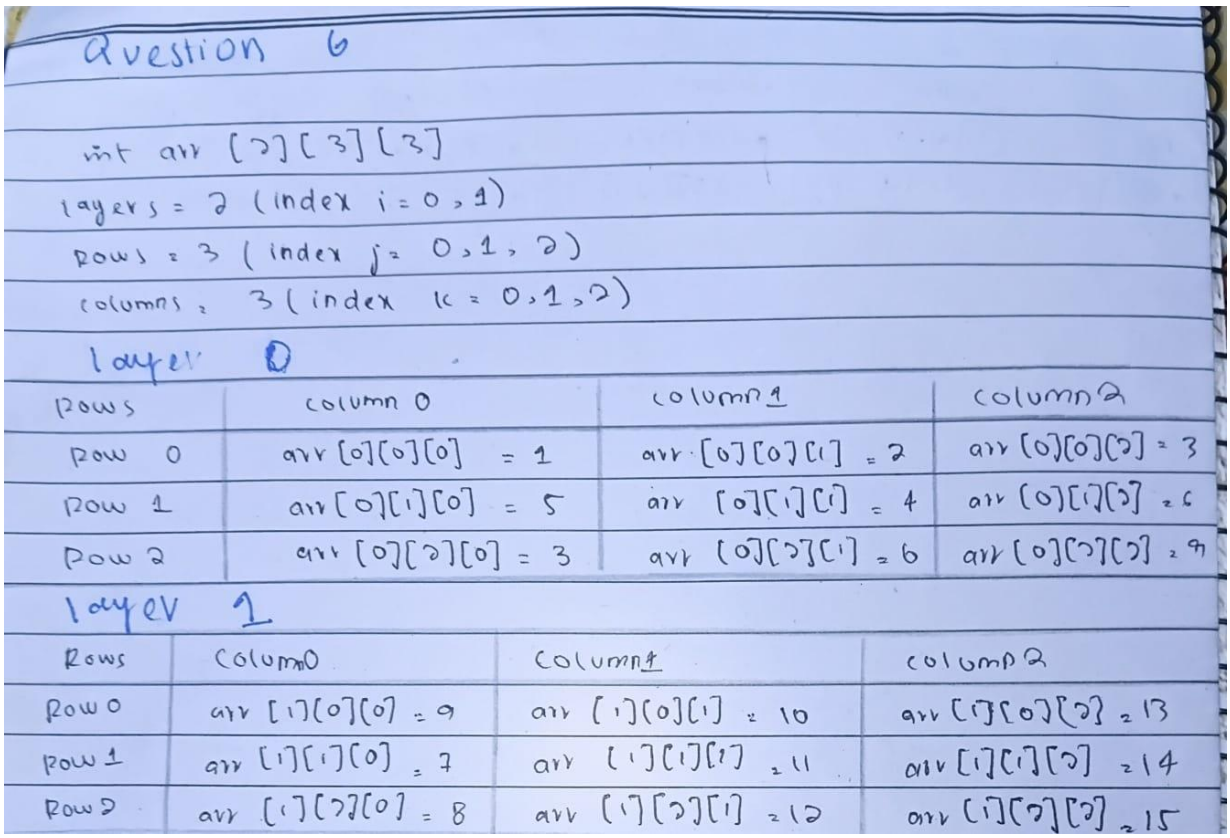
student	subject 1	subject 2	subject 3
student 1	80	75	67
student 2	85	78	71
student 3	83	79	76

subject 1	subject 2	subject 3
marks [1][0][0] 80	marks [1][0][1] 75	marks [1][0][2] 67
marks [1][1][0] 85	marks [1][1][1] 78	marks [1][1][2] 71
marks [1][2][0] 83	marks [1][2][1] 79	marks [1][2][2] 76

PAK COPY HOUSE

Question 6: 3D Array on Paper (Paper Work)

On paper, draw the 3D array `int arr[2][3][3]` as 2 layers (pages), each layer containing a 3×3 grid. Label every cell with its index in the form `arr[i][j][k]` and label the layers, rows and columns. Fill the cells with any values of your choice.



Question 6

int arr [2][3][3]

layers = 2 (index i = 0, 1)

Rows = 3 (index j = 0, 1, 2)

columns = 3 (index k = 0, 1, 2)

layer 0

Rows	column 0	column 1	column 2
Row 0	arr[0][0][0] = 1	arr[0][0][1] = 2	arr[0][0][2] = 3
Row 1	arr[0][1][0] = 5	arr[0][1][1] = 4	arr[0][1][2] = 6
Row 2	arr[0][2][0] = 3	arr[0][2][1] = 6	arr[0][2][2] = 9

layer 1

Rows	column 0	column 1	column 2
Row 0	arr[1][0][0] = 9	arr[1][0][1] = 10	arr[1][0][2] = 13
Row 1	arr[1][1][0] = 7	arr[1][1][1] = 11	arr[1][1][2] = 14
Row 2	arr[1][2][0] = 8	arr[1][2][1] = 12	arr[1][2][2] = 15

Question 7: 3D Array on Paper (Paper Work)

On paper, draw the 3D array `int arr[3][2][4]` as 3 layers (pages), each layer containing 2 rows and 4 columns. Show the index of every cell, fill the array with values, and then write the index of the first and last element of each layer and the total number of elements in the array.

Question 7 :-

int arr [3][2][4]

layer = 3 (index i = 0, 1, 2)

Rows = 2 (index j = 0, 1)

columns 4 (index k = 0, 1, 2, 3)

layer 0

Rows	column 0	column 1	column 2	column 3
Row 0	arr [0][0][0] = 1	arr [0][0][1] = 3	arr [0][0][2] = 4	arr [0][0][3] = 7
Row 1	arr [0][1][0] = 2	arr [0][1][1] = 5	arr [0][1][2] = 6	arr [0][1][3] = 9

layer 1

Rows	column 0	column 1	column 2	column 3
Row 0	arr [1][0][0] = 11	arr [1][0][1] = 12	arr [1][0][2] = 14	arr [1][0][3] = 21
Row 1	arr [1][1][0] = 21	arr [1][1][1] = 13	arr [1][1][2] = 20	arr [1][1][3] = 18

Date _____

layer 2

Rows	column 0	column 1	column 2	column 3
Row 0	arr [2][0][0] = 19	arr [2][0][1] = 17	arr [2][0][2] = 18	arr [2][0][3] = 21
Row 2	arr [2][1][0] = 23	arr [2][1][1] = 21	arr [2][1][2] = 22	arr [2][1][3] = 24